

Pricing and hedging of financial derivatives — Peter Tankov 2023–2024, main session

No documents/calculators allowed, except for 1 sheet of paper (A4, two-sided) with handwritten notes. Exam length 2 hours.

Exercise 1: Employee Stock Ownership Plan (10 points)

An Employee Stock Ownership Plan (ESOP) grants the right to purchase a share of the company at a minimum price between either the initial date T_0 or the expiration date T_1 where $T_0 < T_1$. Additionally, there is a discount of $(1 - \rho)\%$ at the time of purchase. The pay-off of this option at T_1 is then given by

$$H' = S_{T_1} - \rho \min\{S_{T_1}, S_{T_0}\}.$$

1. Show that the price of this option is related in a simple way to the price of the option with pay-off

$$H = \min\{S_{T_1}, S_{T_0}\}.$$

From now on, we focus on the price of this option. We place ourselves in the context of the standard Black-Scholes model with a single risky asset S with constant volatility σ and a risk free asset with constant interest rate r .

2. Introduce a new asset S^0 with value

$$S_t^0 = \begin{cases} S_t, & 0 \leq t \leq T_0 \\ S_{T_0} e^{r(t-T_0)}, & T_0 \leq t \leq T_1. \end{cases}$$

This asset corresponds to holding one unit of S , which is converted into the risk-free asset at time T_0 .

Compute the dynamics of S_0 and show that it follows the Black-Scholes model on the whole interval $[0, T_1]$ with time-dependent deterministic volatility $\sigma_0(t)$ to be determined.

3. Show that

$$H = \min\{S_{T_1}, K S_{T_1}^0\}$$

for some strike K to be determined. Write down the formula for the price at time $t \leq T_0$ of the option with pay-off H , denoted by P_t , involving risk-neutral expectation.

4. We shall now compute P_t using S^0 as numéraire.

- i. Recall the definition of the martingale measure $\mathbb{Q}^0$ associated to the numéraire S^0 .
- ii. Write down the dynamics of the risky asset expressed in the new numéraire $Z_t := \frac{S_t}{S_t^0}$ under the measure $\mathbb{Q}^0$.
- iii. Write down the formula for P_t (for $0 \leq t \leq T_0$) involving expectation under $\mathbb{Q}^0$.
- iv. From the previous point, using the Black-Scholes formula, deduce the explicit expression for the price P_t at time $t \leq T_0$. Hint: $\min(Z, K) = Z - (Z - K)^+$.
- v. Compute the hedging strategy for the option at time $t \leq T_0$.

Exercise 2: Black-Scholes model with default (10 points)

We place ourselves directly under the risk-neutral measure $\mathbb{Q}$. We consider the following extension to the standard Black-Scholes model: at a random time τ , the company defaults: its share price drops down to zero, where it remains until the end. We assume that τ is independent from the Brownian motion W and follows the exponential distribution:

$$\mathbb{P}^{\mathbb{Q}}[\tau > t] = e^{-\lambda t}.$$

The share price dynamics under $\mathbb{Q}$ is thus given by

$$S_t = S_0 e^{(\delta - \sigma^2/2)t + \sigma W_t} \mathbf{1}_{\tau > t}.$$

We also assume that in the market there is a risk-free asset with interest rate r .

1. By computing risk-neutral expectation, show that necessarily, $\delta = r + \lambda$.
2. We now place ourselves at date $t = 0$ and consider a call option on S with maturity T . What is the value of the pay-off of this call option if the company defaults before time T ? Compute the price of the call option, and show that it is equal to the Black-Scholes price, where the interest rate r is replaced with $\delta = r + \lambda$. We denote this price by $\hat{C}(T, K)$.
3. Show that $\hat{C}(T, K)$ is increasing in λ . What is the intuition behind this result?
4. Using Dupire's formula, and the explicit expression of the greeks in the Black-Scholes model, compute the local volatility in this model.
5. Recall the formula from the course: the at the money forward (ATMF) implied volatility $I(T, 0)$ satisfies

$$I(T, 0) \sim \frac{C(T, 0)}{S_0} \sqrt{\frac{2\pi}{T}} \quad \text{as } T \rightarrow 0,$$

where $C(T, 0)$ is the ATMF option price (for $K = S_0 e^{rT}$) in a given model. Use this formula to show that the short-term ATMF volatility in the Black-Scholes model with default equals σ .

6. Recall the formula from the course: the ATMF implied volatility skew satisfies

$$\partial_K I(T, 0) \sim \frac{\sqrt{2\pi}}{\sqrt{\tau}} \{ \mathbb{P}^{BS}[S_T \geq S_0 e^{rT}] - \mathbb{P}^{\mathbb{Q}}[S_T \geq S_0 e^{rT}] \} \quad \text{as } T \rightarrow 0,$$

where under $\mathbb{P}^{BS}$, S follows the standard Black-Scholes model with volatility σ and under $\mathbb{Q}$, S follows the Black-Scholes model with default. Use this formula to compute the implied volatility skew in the Black-Scholes model with default.

7. The model of this exercise does not allow perfect hedging due to the presence of the additional risk factor (default). However, we shall see that the Black-Scholes model allows to obtain a sub-hedging strategy and a lower price bound.
 - i. Show that on the event $\{\tau > T\}$ (no default before time T), the Black-Scholes delta-hedging strategy provides a perfect hedge for the call option with pay-off $(S_T - K)^+$.
 - ii. Assume now that the default happens at some time $\tau \leq T$. What is the value of the Black-Scholes hedging portfolio just before default? What is the value of the Black-Scholes hedging portfolio just after default? Deduce that the Black-Scholes hedging strategy is a sub-hedging strategy for the option in all cases (both when $\tau \leq T$ and when $\tau > T$). Deduce a lower bound on the price of the option.

Solution

Exercise 1: Employee Stock Ownership Plan (10 points)

1. Let $P(t, S)$ be the price of the option with pay-off H and $P'(t, S)$ the price of the option with pay-off H' . Then, $P'(t, S) = S_t - \rho P(t, S)$.

2. We have:

$$\frac{dS_t^0}{S_t^0} = r dt + \sigma_0(t) dW_t$$

with $\sigma_0(t) = \sigma \mathbf{1}_{t \leq T_0}$.

3. $K = e^{-r(T_1 - T_0)}$. Then,

$$P_t = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T_1 - t)} \min\{S_{T_1}, K S_{T_1}^0\} | \mathcal{F}_t \right].$$

4. i. The density of the change of measure is given by

$$\frac{d\mathbb{Q}^0}{d\mathbb{Q}} = e^{-rT_1} \frac{S_{T_1}^0}{S_0} = e^{-\frac{1}{2} \int_0^{T_1} \sigma_0^2(t) dt + \int_0^{T_1} \sigma_0(t) dW_t} = e^{-\frac{\sigma^2}{2} T_0 + \sigma W_{T_0}}$$

- ii. Under the measure $\mathbb{Q}^0$, the process $Z_t = \frac{S_t}{S_t^0}$ is a martingale and satisfies

$$\frac{dZ_t}{Z_t} = (\sigma - \sigma_0(t)) dW_t^0.$$

- iii. By the pricing formula under change of numéraire,

$$\begin{aligned} P_t &= S_t^0 \mathbb{E}^{\mathbb{Q}^0} [\min\{Z_{T_1}, K\} | \mathcal{F}_t] = S_t^0 \mathbb{E}^{\mathbb{Q}^0} [Z_{T_1} - (Z_{T_1} - K)^+ | \mathcal{F}_t] \\ &= S_t - S_t^0 \mathbb{E}^{\mathbb{Q}^0} [(Z_{T_1} - K)^+ | \mathcal{F}_t]. \end{aligned}$$

- iv. By the Black-Scholes formula,

$$P_t = S_t - S_t N(d_1) + S_t K N(d_2)$$

with

$$d_{1,2} = \frac{\log(Z_t/K) \pm \frac{1}{2} \int_t^{T_1} (\sigma - \sigma_0(s))^2 ds}{\sqrt{\int_t^{T_1} (\sigma - \sigma_0(s))^2 ds}} = \frac{r(T_1 - T_0) \pm \frac{\sigma^2}{2} (T_1 - T_0)}{\sigma \sqrt{T_1 - T_0}}$$

- v. Since P_t depends linearly on S_t , the hedge ratio before time T_0 is constant and equal to

$$\delta_t = 1 - N(d_1) + K N(d_2)$$

Exercise 2: Black-Scholes model with default (10 points)

1. By risk-neutral expectation,

$$S_0 = \mathbb{E}^{\mathbb{Q}}[e^{-rt} S_t] = \mathbb{E}^{\mathbb{Q}}[e^{-rt} S_0 e^{(\delta - \sigma^2/2)t + \sigma W_t} \mathbf{1}_{\tau > t}] = S_0 \mathbb{E}^{\mathbb{Q}}[e^{-rt} S_0 e^{(\delta - \sigma^2/2)t + \sigma W_t}] \mathbb{E}^{\mathbb{Q}}[\mathbf{1}_{\tau > t}] = S_0 e^{(\delta - r - \lambda)t}.$$

Thus, $\delta = r + \lambda$.

2. If the company defaults before time t , the pay-off is zero. By risk-neutral expectation, we then have:

$$\begin{aligned} \hat{C}(T, K) &= \mathbb{E}^{\mathbb{Q}}[e^{-rT} (S_T - K)^+] = \mathbb{E}^{\mathbb{Q}}[e^{-rT} (S_0 e^{(\delta - \sigma^2/2)T + \sigma W_T} - K)^+ \mathbf{1}_{\tau > T}] \\ &= \mathbb{E}^{\mathbb{Q}}[e^{-(r+\lambda)T} (S_0 e^{(r+\lambda - \sigma^2/2)T + \sigma W_T} - K)^+] \\ &= S_0 N(d_1) - K e^{-(r+\lambda)T} N(d_2), \quad d_{12} = \frac{\log \frac{S_0}{K e^{-(r+\lambda)T}} \pm \frac{\sigma^2}{2} T}{\sigma \sqrt{T}}. \end{aligned}$$

We recognize the expression of the Black-Scholes price with r replaced by $r + \lambda$.

3. To show that the price is increasing in λ , rewrite it as follows:

$$\widehat{C}(T, K) = \mathbb{E}^{\mathbb{Q}}[(S_0 e^{-\sigma^2 T/2 + \sigma W_T} - K e^{-(r+\lambda)T})^+].$$

Since the expectation under the integral sign is increasing, the same is true for the option price. The higher λ , the more risky is the stock, and the higher the price of the protection.

4. By Dupire's formula,

$$\sigma^2(T, K) = 2 \frac{\frac{\partial \widehat{C}}{\partial T} + rK \frac{\partial \widehat{C}}{\partial K}}{K^2 \frac{\partial^2 \widehat{C}}{\partial K^2}}$$

To simplify this expression, we can use the fact that

$$\sigma^2 = 2 \frac{\frac{\partial \widehat{C}}{\partial T} + (r + \lambda)K \frac{\partial \widehat{C}}{\partial K}}{K^2 \frac{\partial^2 \widehat{C}}{\partial K^2}}.$$

Therefore,

$$\sigma^2(T, K) = \sigma^2 - 2 \frac{\lambda \frac{\partial \widehat{C}}{\partial K}}{K \frac{\partial^2 \widehat{C}}{\partial K^2}}$$

Let us compute the derivatives with respect to K :

$$\frac{\partial \widehat{C}}{\partial K} = -e^{-(r+\lambda)T} N(d_2), \quad \frac{\partial^2 \widehat{C}}{\partial K^2} = \frac{e^{-(r+\lambda)T} n(d_2)}{K \sigma \sqrt{T}},$$

so that

$$\sigma^2(T, K) = \sigma^2 + 2\lambda \frac{N(d_2) \sigma \sqrt{T}}{n(d_2)}$$

5. The ATMF option price in the Black-Scholes model with default is given by

$$C(T, 0) = S_0 N(d_1) - S_0 e^{-\lambda T} N(d_2), \quad d_{12} = \frac{\lambda \pm \frac{\sigma^2}{2}}{\sigma} \sqrt{T}.$$

Taylor expansion yields: $C(T, 0) = S_0 \sigma \sqrt{T} + O(T)$. Comparing this with the formula for the implied volatility given in the statement, we conclude that $I(T, 0) \sim \sigma$ as $T \rightarrow 0$.

6. We compute:

$$\begin{aligned} \mathbb{P}^{BS}[S_T \geq S_0 e^{rT}] &= \mathbb{P}[\sigma W_T - \frac{\sigma^2}{2} T \geq 0] = N(-\sigma \sqrt{T}/2) = \frac{1}{2} - \frac{\sigma \sqrt{T}}{2\sqrt{2\pi}} + o(T) \\ \mathbb{P}^{\mathbb{Q}}[e^{(\lambda - \sigma^2/2)T + \sigma W_T} \mathbf{1}_{\tau > T} \geq 1] &= e^{-\lambda T} \mathbb{P}^{\mathbb{Q}}[(\lambda - \sigma^2/2)T + \sigma W_T \geq 0] = \frac{1}{2} - \left(\frac{\sigma}{2} - \frac{\lambda}{\sigma}\right) \frac{\sqrt{T}}{\sqrt{2\pi}} + o(T) \end{aligned}$$

Substituting these expressions into the formulas for the implied volatility skew, we get:

$$\partial_k I(T, 0) = -\frac{\lambda}{\sigma}.$$

7. i. On the event $\{\tau > T\}$ the share price follows the standard Black-Scholes model, hence the Black-Scholes hedging strategy provides the perfect hedge.
ii. The Black-Scholes hedging portfolio contains $\delta_t = N(d_t^1)$ units of stock and a position in the risk-free asset of size $-K e^{-r(T-t)} N(d_t^2)$ (negative amount). Just after default, the stock position drops to zero, and the value of the portfolio becomes equal to

$$V_\tau = -K e^{-r(T-\tau)} N(d_\tau^2).$$

Assuming that no transactions are made after default, the value of the portfolio at maturity is $V_T = -K N(d_T^2) < 0$, while the pay-off value is zero. Hence the Black-Scholes strategy is a sub-hedging strategy, and the Black-Scholes price is a lower bound on the price.